

Introduction

Gastric cancer (GC) is one of the most common cancer types and the third leading cause of cancer-related deaths worldwide. GC is a complex disease characterized by the involvement of multiple processes and factors. Correa's cascade describes the multi-step progression from nonactive or chronic active gastritis to gastric precancerous stages, including atrophic gastritis and intestinal metaplasia, eventually resulting in gastric adenocarcinoma. Given that the frequent diagnosis of GC at the advanced stage leads to a high mortality rate, it is crucial to develop strategies to treat the early or premalignant stages of GC. However, the cellular mechanisms involved in the transition from initial tumor development to early gastric cancer (EGC) remain poorly understood.

Tumor progression is influenced by the tumor microenvironment (TME), which includes the extracellular matrix, stromal cells such as fibroblasts, and tumor-associated immune cells, through autocrine and paracrine cytokines or other signaling factors. Single-cell RNA sequencing (scRNA-seq) has proven to be an effective approach for investigating the TME by profiling gene expression in individual cells.

In this study, we performed single-cell transcriptome analysis using publicly available datasets to unravel the intricate cellular composition of the TME throughout the progression of gastric tumors. We analyzed CD8⁺ T cells and fibroblasts in the TME across various disease stages, as well as latent intercellular interactions between CD8⁺ T cells and epithelial cells, to better understand the biological systems of human gastric tumors and to advance novel therapeutic methods.

Methods

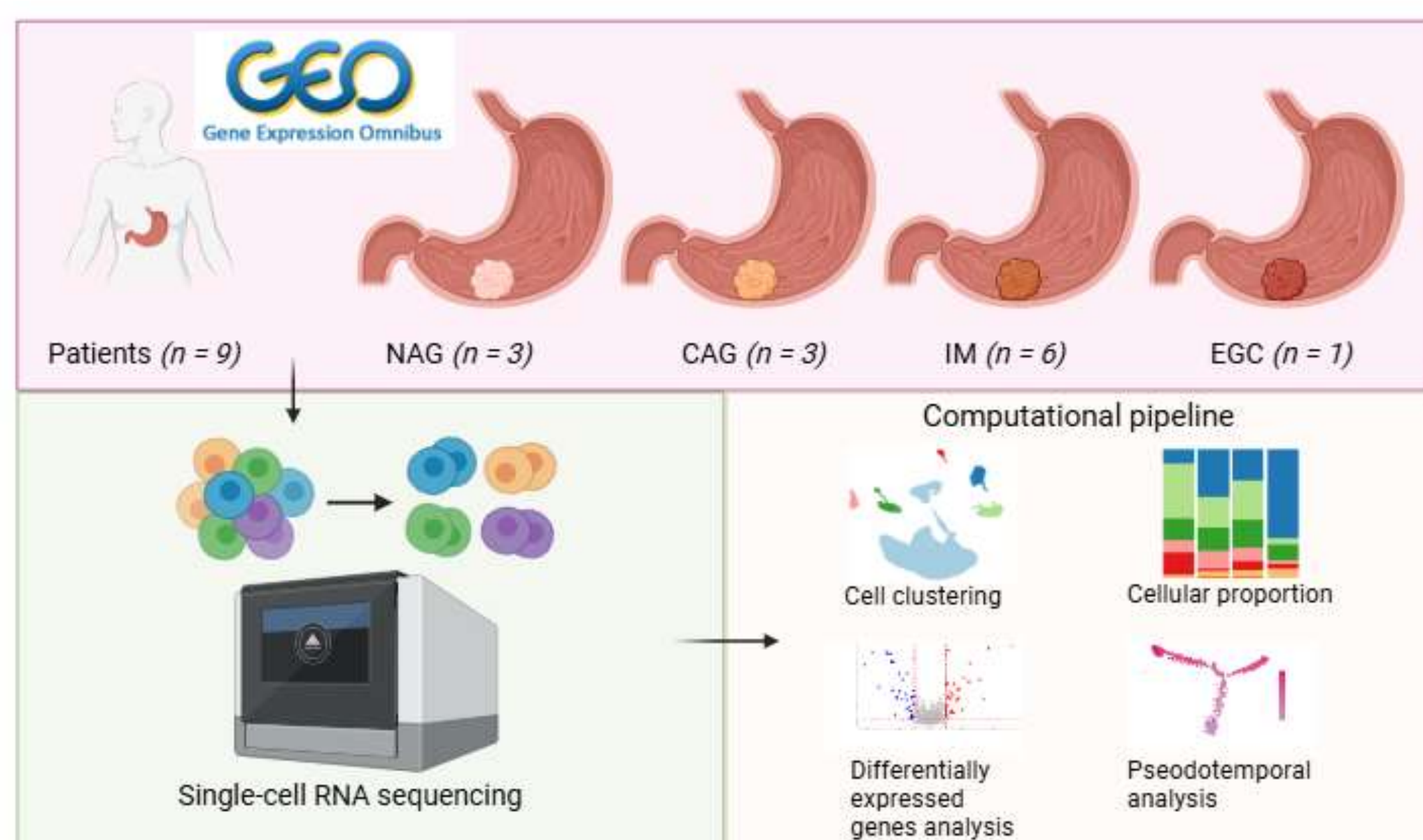


Figure 1. Schematic illustration of the study design

All 13 samples were analyzed to depict the single-cell landscape of gastric tumors. The illustration was created with BioRender (<https://www.biorender.com/>).

NAG, non-atrophic gastritis; CAG, chronic atrophic gastritis; IM, intestinal metaplasia; EGC, early gastric cancer.

Results

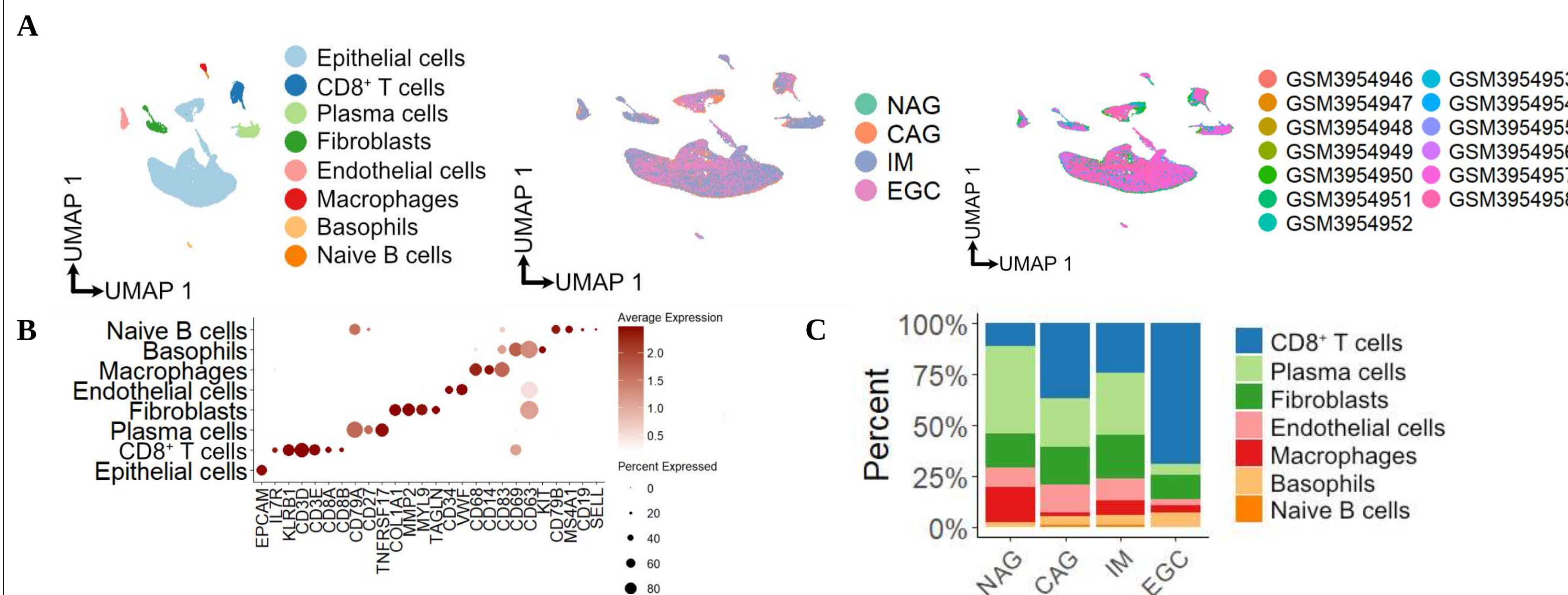


Figure 2. Single-cell landscape of cellular components in gastric tumors according to Correa's cascade.

(A) UMAP visualization of 44,753 cells passed quality control from 13 samples. Cell types, disease stages, and samples are distinguished by different colors.

(B) Expression of canonical cell type marker genes. The average expression is shown by the color intensity, while the dot size represents the percentage of cells expressing the representative markers.

(C) Bar plot showing the proportion of cell types across different disease stages. Each bar is matched to disease types, and colors are matched to cell types.

UMAP, uniform manifold approximation and projection; NAG, non-atrophic gastritis; CAG, chronic atrophic gastritis; IM, intestinal metaplasia; EGC, early gastric cancer.

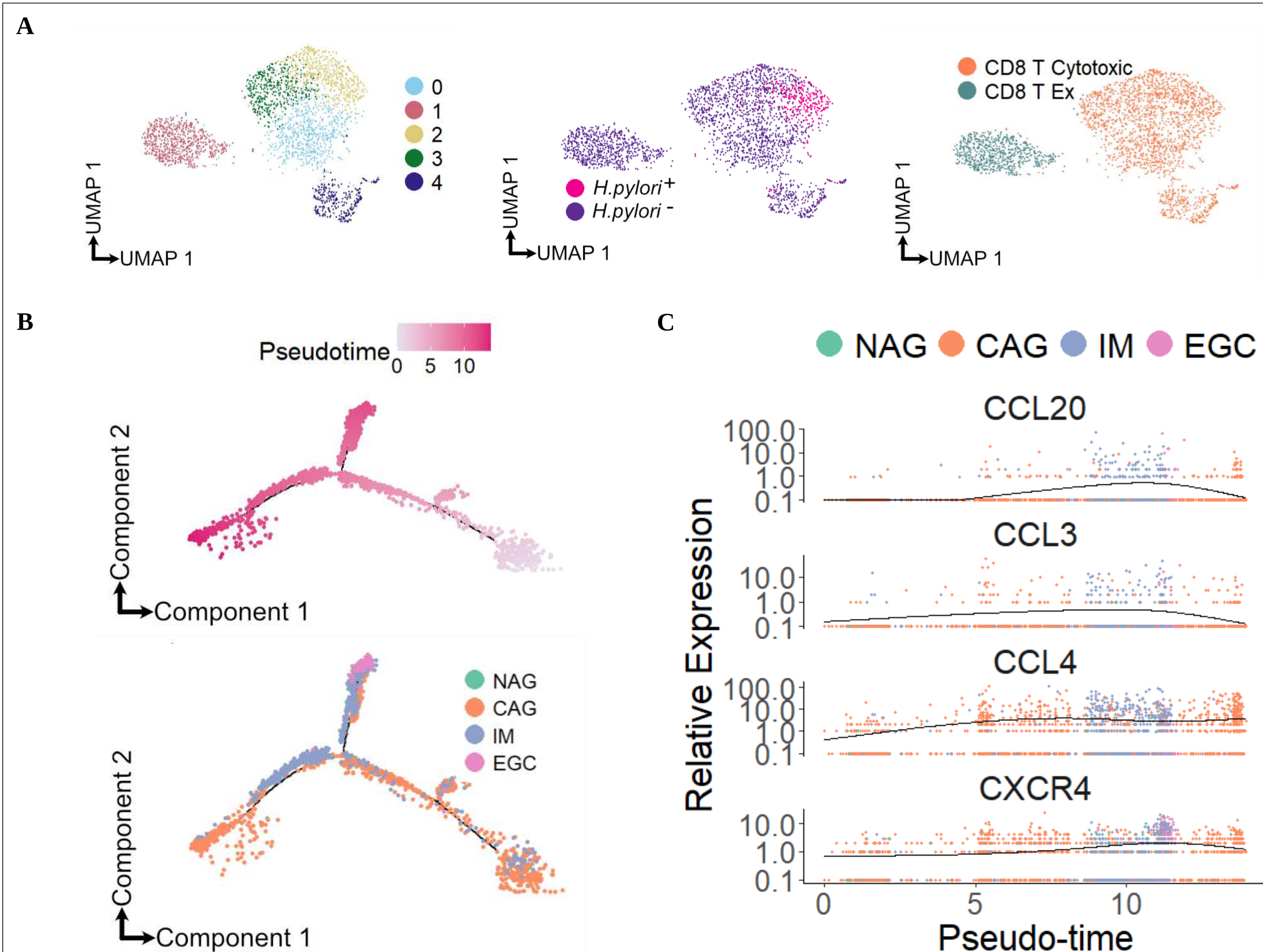


Figure 3. Characterization of CD8⁺ T cells and their developmental trajectories.

(A) UMAP visualization of 2665 CD8⁺ T cells. Colors are coded across subclusters, *H. pylori* infection, and functional phenotypes of CD8⁺ T cells.

(B) Developmental trajectory of CD8⁺ T cells. Colors are coded across disease stages and pseudotime.

(C) Gene expression levels of *CCL20*, *CCL3*, *CCL4*, and *CXCR4* in CD8⁺ T cells along the pseudotime. The X-axis indicates pseudotime and the Y-axis shows relative expression.

UMAP, uniform manifold approximation, and projection; NAG, non-atrophic gastritis; CAG, chronic atrophic gastritis; IM, intestinal metaplasia; EGC, early gastric cancer; *H. pylori*, *Helicobacter pylori*.

Results

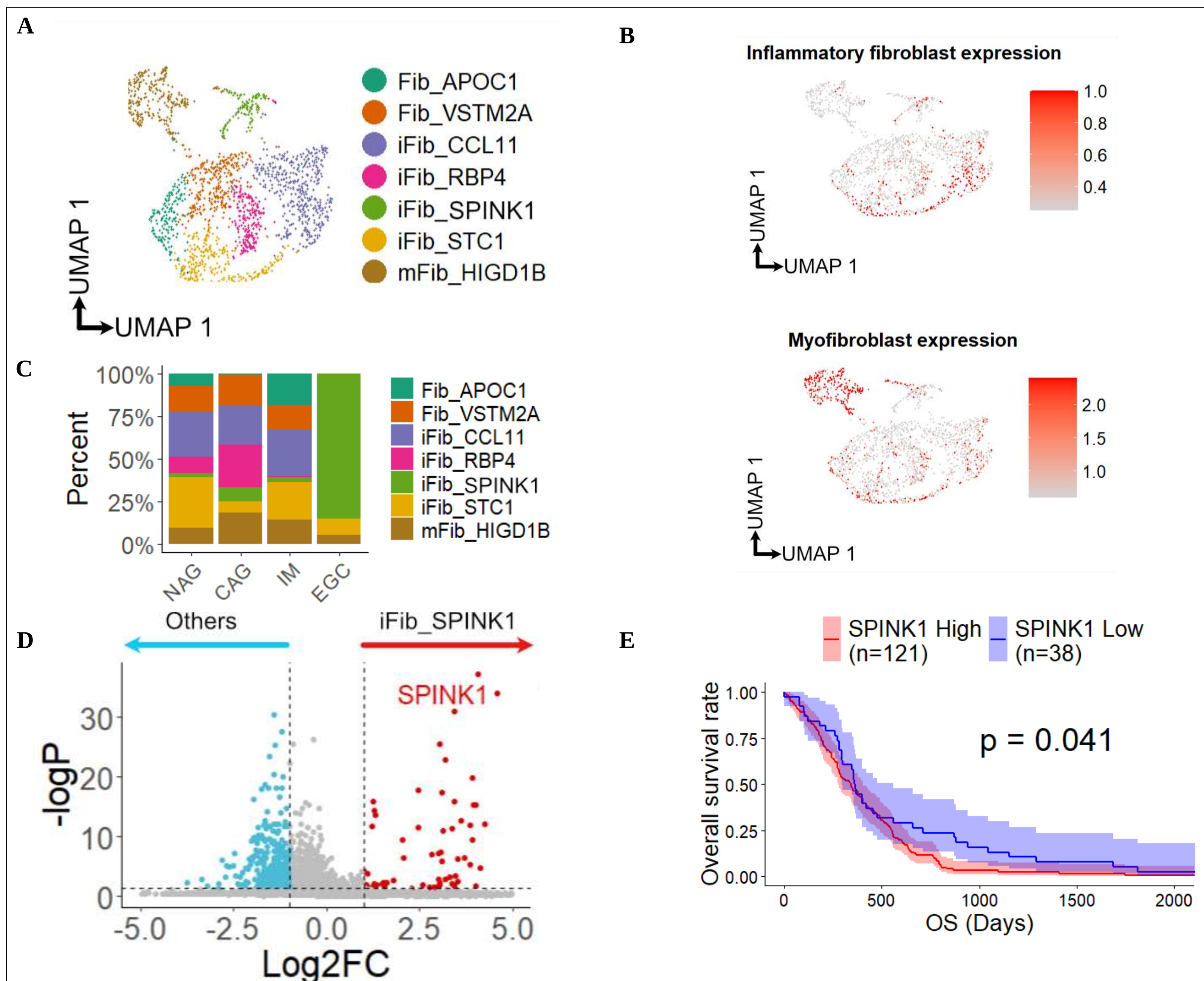


Figure 4. Characterization of fibroblasts during gastric tumor development.

(A) UMAP visualization of 1792 fibroblasts. Colors are coded across fibroblast subclusters.

(B) UMAP representing the average expression of inflammatory fibroblast markers (*CXCL1*, *CXCL2*, *MMP1*, *MMP3*, *CXCL12*) and myofibroblast markers (*ACTA2*, *TPM1*, *TPM2*, *MYL9*, *TAGLN*).

(C) Bar plot showing the proportion of fibroblasts across different disease stages. Each bar is matched to disease types, and colors are matched to fibroblast subclusters.

(D) Volcano plot of differentially expressed genes between iFib_SPINK1 and other fibroblasts. Significant genes are colored by red and blue dots (P-value < 0.05 and |log₂ fold change| > 1). The X-axis indicates the log₂ fold change, and the Y-axis shows the -log P value.

(E) Kaplan-Meier plots showing the overall survival of patients with stomach adenocarcinoma in the TCGA-STAD dataset divided into the high and low expression of *SPINK1*. The statistical significance was calculated by the Log-rank test, P = 0.041.

UMAP, uniform manifold approximation and projection; iFib, inflammatory fibroblast; mFib, myofibroblast; NAG, non-atrophic gastritis; CAG, chronic atrophic gastritis; IM, intestinal metaplasia; EGC, early gastric cancer; FC, fold change; TCGA-STAD, The cancer genome atlas-stomach adenocarcinoma.

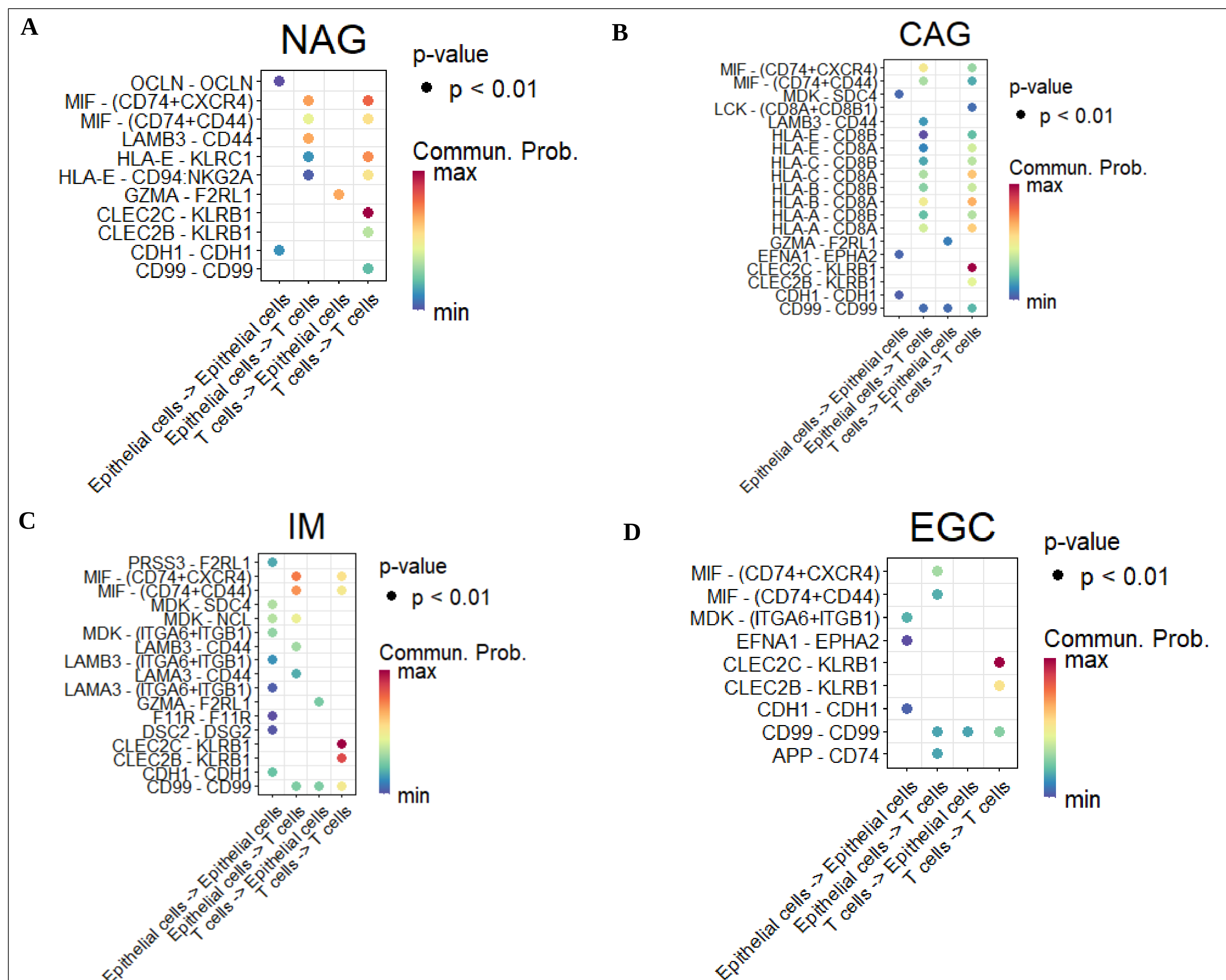


Figure 5. Intercellular interaction analysis in gastric tumors.

(A-D) Bubble plots showing disparate ligand-receptor interactions between epithelial cells and CD8⁺ T cells in NAG, CAG, IM, and EGC. Colors display the communication probability of the ligand and receptor pairs, and the circle size indicates the statistical significance.

NAG, non-atrophic gastritis; CAG, chronic atrophic gastritis; IM, intestinal metaplasia; EGC, early gastric cancer.

Conclusion

Our results showed an increase in the infiltration of CD8⁺ T cells and a decrease in the proportion of plasma cells during gastric tumor development. However, fibroblasts maintained a constant presence as gastric tumors progressed.

Subpopulation analyses of CD8⁺ T cells and fibroblasts revealed an enrichment of exhausted CD8⁺ T cells in *Helicobacter pylori* (*H. pylori*) infection-negative gastric tumors. We also found that gene expression patterns related to inflammatory responses, including *CCL20*, *CCL3*, *CCL4*, and *CXCR4*, gradually increased along their developmental trajectories, especially in an EGC branch. The subpopulations of fibroblasts showed dynamic changes in composition and *SPINK1*⁺ fibroblasts were enriched in EGC.

In our study, we uncovered the overall intercellular interactions between CD8⁺ T cells and epithelial cells mediated by ligand-receptor communications in the TME. We identified that the APP-CD74 axis, which could promote immunosuppression and progression of cancer, was significantly represented in the EGC stage.

To solidify our findings, future research should confirm these results through experimental and clinical assessments of cell populations and their prognostic significance in patients.

References

- [1] Liu, Qingsong, et al. Berberine for gastric cancer prevention and treatment: Multi-step actions on the Correa's cascade underlie its therapeutic effects. *Pharmacological Research*, 2022, 184: 106440.
- [2] Wang, Ruiping, et al. Evolution of immune and stromal cell states and ecotypes during gastric adenocarcinoma progression. *Cancer cell*, 2023, 41.8: 1407-1426.
- [3] Li, Wenke, et al. Unveiling promising targets in gastric cancer therapy: A comprehensive review. *Molecular Therapy Oncology*, 2024.